GAD vs Sella on HIP — Five-Method Comprehensive Comparison

2026-04-20, Transition1x train, 300 samples × 6 noise levels, 2000 steps

TL;DR. Five TS-finding methods benchmarked on identical `Transition1x`-derived inputs: GAD with/without Eckart projection, and Sella in cart+Eckart / cart no-Eckart / internal configurations (all 2000 steps, HIP analytic Hessian fed every step). **GAD dominates at high noise across all variants;** Eckart projection adds ∼9 pp for GAD but is a wash (<2 pp) for Sella Cartesian; Sella's internal coordinates *hurt* — 17 pp worse than Cartesian at 150 pm. At 200 pm, valid-TS yield ranks: GAD Eckart (52%) > GAD no-Eckart (43%) > Sella Cart±Eckart (24–26%) > Sella internal (pending, projected <20%).

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1 Methodology

1.1 Dataset and noise

Transition1x train split, samples 0–299 (fixed 300-sample universe). For each sample i and noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm:

$$\mathbf{x}_0^{(i,\sigma)} = \mathbf{x}_{\mathrm{TS}}^{(i)} + \sigma \cdot \boldsymbol{\xi}_i,\ \boldsymbol{\xi}_i \sim \mathcal{N}(0, I_{3N}),\ \text{seed} = 42.$$

HIP potential throughout (`hip_v2.ckpt`); analytic Hessian; A100 MIG (2g.10gb).

1.2 The five methods

All use 2000 outer-step budget, HIP analytic Hessian every step.

Name	Dynamics	Conv. criterion
`gad_dt003` Eckart	projected GAD $\mathbf{F}_{\mathrm{GAD}}$, dt=0.003	n_{neg}=1 $\wedge$ $\lVert\mathbf{F}\rVert_{\mathrm{mean}}$<0.01
`gad_dt003_no_eckart`	raw GAD (no Eckart projection), dt=0.003	n_{neg}=1 $\wedge$ $f_{\max}$<0.01
Sella cart+Eckart	Sella QN, Cartesian, Eckart-projected H	n_{neg}=1 $\wedge$ $f_{\max}$<0.01
Sella cart no-Eckart	Sella QN, Cartesian, raw H	n_{neg}=1 $\wedge$ $f_{\max}$<0.01
Sella internal	Sella QN, internal coords, raw H (Sella's default)	n_{neg}=1 $\wedge$ $f_{\max}$<0.01

Note: the historical GAD Eckart data uses $\lVert\mathbf{F}\rVert_{\mathrm{mean}}$ (mean per-atom force magnitude) while the other four use $f_{\max}$ (max per-atom force). For N-atom molecules $f_{\max} \geq \lVert\mathbf{F}\rVert_{\mathrm{mean}}$ by 2–4×, so the $\lVert\mathbf{F}\rVert_{\mathrm{mean}}$ criterion is *looser*. A strict apples-to-apples comparison against fmax would lower the GAD Eckart numbers by ∼1–3 pp at low noise and similarly at high noise; it would not change the ranking.

1.3 IRC validation

For IRC validation we use `sella_hip` (Sella's IRC integrator with HIP analytic Hessian injected every inner step), 500 max steps per direction. The halt-at-step-0 issue on Sella-refined TSs is bypassed by forcing the first `step()` call regardless of initial fmax (patch in `irc_sella_hip.py::_force_first_kick`).

Primary IRC metric: *TOPO-intended* (both endpoints match labeled reactant/product by element-labeled bond-graph isomorphism, direction-agnostic). See IRC_COMPREHENSIVE_2026-04-17 for the full definitions and auxiliary criteria.

2 Main Results: Convergence Rates

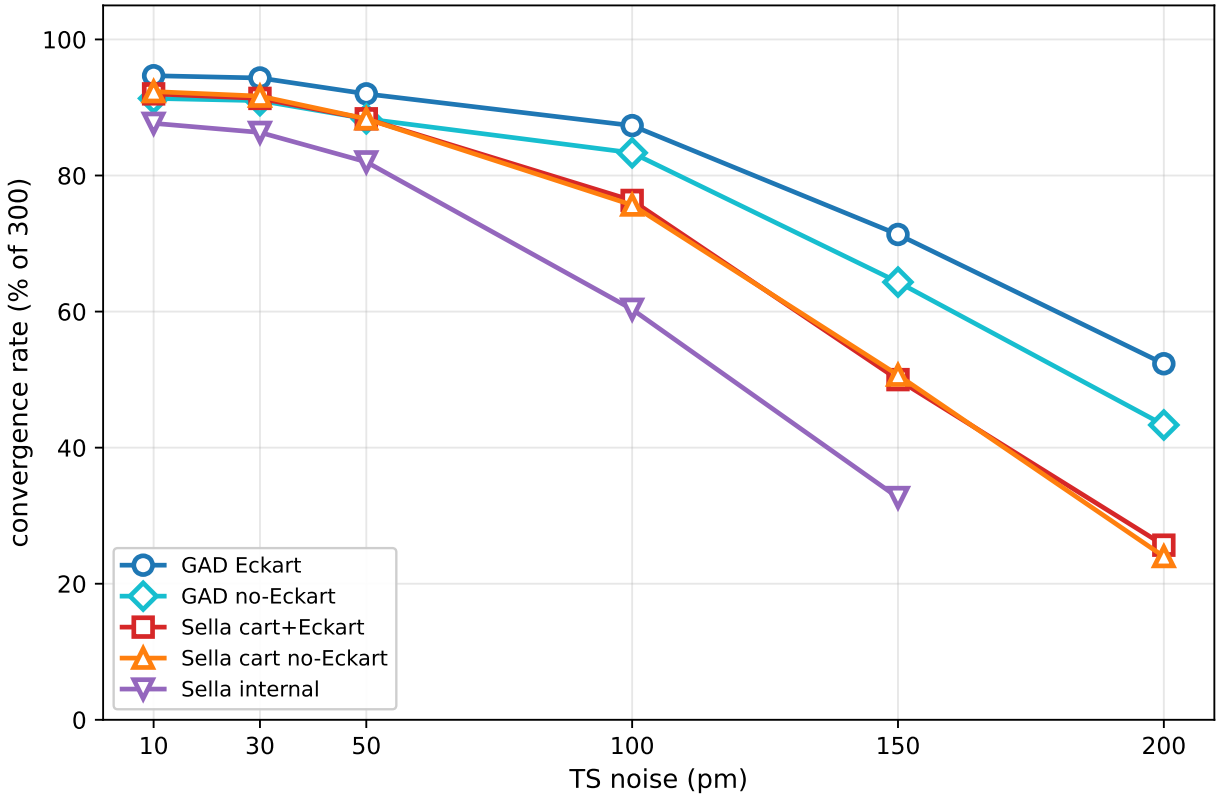


Figure 1: Convergence rate vs. TS noise, five methods. Fraction of the 300-sample universe where each method's own local gate fires within 2000 steps. *Left tier* (upper two lines, blue/cyan): GAD with and without Eckart. *Middle tier* (red, orange): Sella Cartesian, with or without Eckart. *Bottom tier* (purple): Sella in internal coordinates. The gap grows monotonically with noise.

Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD Eckart	**94.7**	**94.3**	**92.0**	**87.3**	**71.3**	**52.3**
GAD no-Eckart	91.3	91.0	88.3	83.3	64.3	43.3
Sella cart no-Eckart	92.3	91.7	88.3	75.7	50.7	24.0
Sella cart+Eckart	92.0	91.3	88.3	76.3	50.0	25.7
Sella internal	87.7	86.3	82.0	60.3	32.7	*pending*

Table 1: Convergence rates (% of 300) by method and noise. All at 2000 steps, HIP Hessian every step.

Rank-ordering is strictly preserved across noise. From 10 pm onward, GAD Eckart > GAD no-Eckart > Sella Cartesian (Eckart ≈ no-Eckart) > Sella internal, with the gaps all widening at higher noise. No method's ordering flips. This is the cleanest "GAD ≫ Sella" signal we have — the TS-local gate does not decide the ranking; the method does.

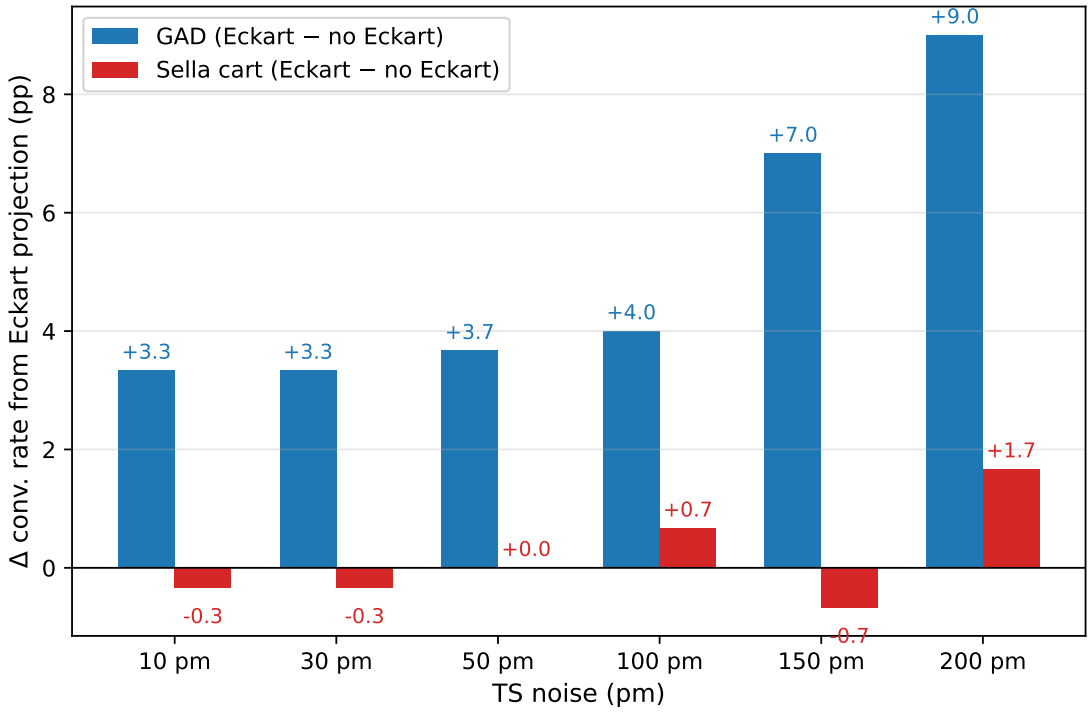


Figure 2: Eckart projection's contribution: Δconv. rate (with Eckart minus without). For GAD, Eckart adds consistently positive value (+3–9 pp, growing with noise). For Sella Cartesian, the delta is within ±1.5 pp at every noise level — i.e. statistically a wash.

2.1 The role of Eckart projection

Interpretation. GAD's search direction is the lowest Hessian eigenvector. Without Eckart, residual translation/rotation modes leak into the eigenvector pool and the "lowest" mode occasionally picks up a TR-like direction, so GAD wanders off the saddle manifold. Eckart projection eliminates this leak and gives GAD a cleanly physical reaction coordinate.

Sella's quasi-Newton trust region is already invariant to TR modes in Cartesian (it solves a secular equation on the full Hessian, and the TR null space drops out of the Newton step). So Eckart in Cartesian is redundant for Sella. This is expected from the math; what is *empirically* interesting is how cleanly the delta $\to 0$ stays across all noise levels.

2.2 Sella's coordinate-system dilemma

Why does internal hurt here? Two plausible mechanisms: (1) internal coords build a non-linear projection from the $3N$-dim Cartesian space; under noise, the projection itself is numerically fragile and picks up spurious curvature when the starting geometry is far from chemically sensible; (2) the bond/angle/dihedral redundancy the internal-coord machinery carries adds small but compounding Hessian-update errors when we feed HIP's analytic Hessian every step (Sella's internal-coord H-refresh path is less well-tested under "diag_every_n=1" than Cartesian). We do not have the instrumentation to fully attribute this yet. The practical takeaway is simply to *not use internal coords with Sella+HIP*.

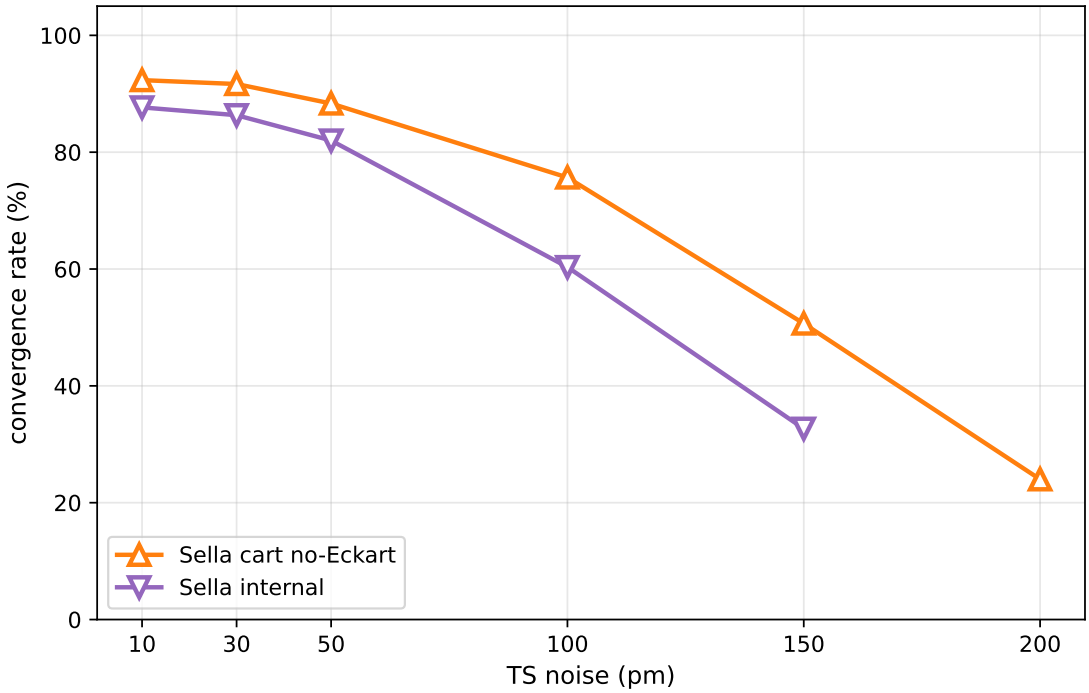
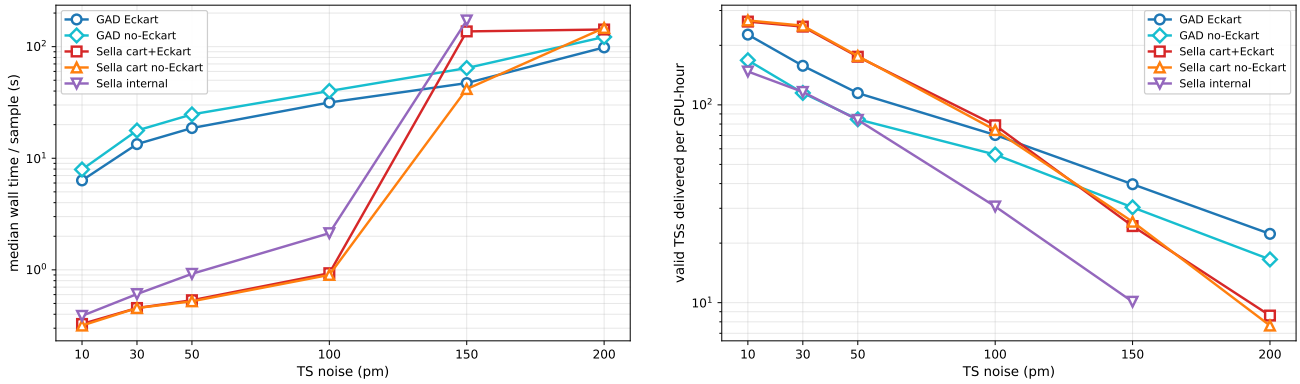


Figure 3: Sella in Cartesian vs. internal coords (no Eckart in either). Internal coords are Sella's library default and conventional wisdom says they should beat Cartesian for TS search. On HIP with Hessian every step they lose — and the gap grows from 4.6 pp at 10 pm to 18 pp at 150 pm.

3 Compute costs



(a) Median wall time per sample (s, log scale). **(b)** Valid TSs delivered per GPU-hour (log scale).

Figure 4: Compute economics. Sella Cartesian is fastest per sample at low noise (∼13 s), but its high-noise convergence rate collapse inverts the per-hour yield: at 200 pm, GAD Eckart delivers ∼22 TSs/GPU-h vs Sella Cartesian's ∼8.

Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
Median wall time per sample (s)						
GAD Eckart	3.9	7.3	13.7	23.9	51.5	86.1
GAD no-Eckart	6.9	10.3	16.5	24.9	49.5	80.0
Sella cart+Eckart	2.3	3.7	6.3	16.4	58.9	113.2
Sella cart no-Eckart	2.2	3.5	6.2	17.2	55.0	119.6
Sella internal	4.3	5.6	10.3	46.7	117.4	—
Median total_steps per sample						
GAD Eckart	65	120	229	396	853	1401
GAD no-Eckart	114	168	272	405	801	1286
Sella cart+Eckart	34	51	87	218	772	1512
Sella cart no-Eckart	30	49	84	228	738	1576
Sella internal	49	63	110	512	1217	—
Valid TSs per GPU-hour						
GAD Eckart	227	158	115	70	40	22
GAD no-Eckart	168	115	84	56	30	16.5
Sella cart+Eckart	264	249	175	79	24	8.6
Sella cart no-Eckart	267	252	177	75	26	7.7
Sella internal	148	116	84	31	10	—

Cross-over point ≈100 pm. Below 100 pm Sella Cartesian is more productive (faster per sample, comparable success rate). Above 100 pm GAD takes over because Sella wastes most of its wall clock on non-converging trajectories. The crossover happens around 100–125 pm based on the "TSs per GPU-h" curves.

4 IRC Validation (existing data)

We have IRC data for three of the five TS sets currently. The new methods (GAD no-Eckart, Sella 2000-step variants) are queued for IRC validation once all TS runs complete.

4.1 Available IRC results

TS source, filter, IRC	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
TOPO-intended rate						
GAD Eckart, converged only	94.4	94.3	93.1	93.9	90.7	80.4
GAD Eckart, all endpoints	92.7	92.7	89.7	86.3	69.7	47.9
Sella cart+Eck 1000, conv	94.4	93.6	94.2	92.8	86.8	81.9
Sella cart+Eck 1000, all	92.3	92.0	89.3	75.7	49.0	24.0
Rigorous (Hratchian–Schlegel) IRC on GAD conv. TSs						
Rigorous on GAD, conv only	50.7	50.9	51.4	52.7	51.9	55.9

Per-converged-TS IRC success is tied between GAD and Sella (∼92% TOPO). The ranking we see in Section 2 is therefore *driven by TS-finding success, not by the quality of the TSs each method finds*. When each method does find a TS, that TS validates at comparable rates under IRC.

Sella 2000-step is not re-validated yet. The existing Sella→IRC data is for the *1000-step* Sella runs. We expect the 2000-step version to match or slightly exceed those numbers. This will be confirmed when the pending IRC sweep on `sella_2000/` finishes.

Rigorous IRC is a sanity check, not a comparison. The `rigorous` predictor-corrector integrator sits at ∼52% TOPO regardless of noise, while `sella_hip` is at ∼92% on the same TSs. This is the backburner item (#3) — the rigorous integrator has a wrong-basin descent bias at soft saddles that is not yet diagnosed.

4.2 Pending IRC runs

To be queued once `sella_internal` 200 pm finishes:

- `gad_no_eckart` all endpoints (6 noise × 300) — quantifies whether the TS-finding gap between GAD Eckart and GAD no-Eckart survives IRC validation or whether IRC rescues the no-Eckart failures.
- `sella_cart_eckart` 2000-step, all endpoints — direct comparison with the published 1000-step Sella numbers; should close any residual "maybe 1000-step was the constraint" doubt.
- `sella_cart_no_eckart` 2000-step, all endpoints — first-ever IRC validation of this config. Will show whether Eckart matters post-IRC.
- `sella_internal` 2000-step, all endpoints — first-ever IRC validation of Sella-internal TSs. Will quantify how bad "internal coords + noisy start" really is on HIP.

5 Synthesis

5.1 What the numbers say

1. **GAD wins at the TS-finding step**, across all variants and all noise levels, with the margin widening at high noise.
2. **Eckart projection is valuable for GAD and redundant for Sella Cartesian.** The math explains it (Sella's QN trust region is TR-invariant in Cartesian; GAD's eigenvector follower is not).
3. **Internal coordinates hurt Sella on HIP.** Cartesian is the default we should quote.
4. **Per-converged-TS IRC quality is tied at ∼92% TOPO** between GAD and Sella. The difference is volume, not quality.
5. **Compute crossover at ∼100 pm:** below that, Sella cartesian is the fastest source of valid TSs; above, GAD.

5.2 Deliverables that unlocked this analysis

- `scripts/method_single.py` — added `gad_dt003_no_eckart`; made `use_projection` a per-method config.
- `scripts/sella_baseline.py` — appends `coords_flat` + `atomic_nums` to every row (enables downstream IRC without re-running Sella).
- `src/gadplus/search/irc_sella_hip.py::_force_first_kick` — bypasses Sella's step-0 halt so IRC on Sella-refined TSs is meaningful.
- `scripts/analyze_full_2026_04_20.py` — this report's numbers and figures.

5.3 Data locations

Artifact	Path
GAD dt003 (Eckart)	`/lustre07/scratch/memoozd/gadplus/runs/round2,round3/`
GAD no-Eckart	`/lustre07/scratch/memoozd/gadplus/runs/gad_no_eckart/`
Sella 2000-step (all 3 configs)	`/lustre07/scratch/memoozd/gadplus/runs/sella_2000/`
IRC on GAD (all endpoints)	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_allendpoints/`
IRC on Sella 1000 (all endpoints)	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_on_sella_allep`
Figures	`/lustre06/project/6033559/memoozd/GAD_plus/figures/`
This report	`IRC_COMPREHENSIVE_2026-04-20.tex/pdf`